

General Knowledge

Indian Geography

STUDY NOTES

Transport - Airways and Pipelines



Airways and Pipelines



Airways in India

Importance of Airways

- **Quickest Mode of Transport:** Airways are the fastest way to travel, especially for long distances and difficult terrains.
- **Strategic Connectivity:** Essential for remote and inaccessible regions like the Northeast, Andaman & Nicobar Islands, and mountainous areas.
- **Boost to Economy:** Promotes tourism, trade, and overall economic growth by connecting major cities and regions.
- **Emergency Services:** Vital for disaster relief, medical emergencies, and defence purposes.

Types of Air Transport

- **Domestic Air Transport:** Operates within the country, connecting different cities and regions. Examples include Indigo, Air India, and SpiceJet.

- **International Air Transport:** Operates flights between India and other countries. Air India and private airlines operate many of these routes.
- **Helicopter Services:** Used for connecting remote regions, as well as for medical and rescue missions.

Key Government Initiatives for Airways

UDAN (Ude Desh ka Aam Nagrik)

- **Objective:** To make air travel affordable for the common man and improve regional connectivity.
- **Features:**
 - Subsidized fares for flights on regional routes.
 - Reduced taxes and financial support for airlines operating on less profitable routes.
- **Impact:** Enhanced connectivity to smaller cities and towns; has made air travel accessible to remote areas and boosted regional tourism and trade.

National Civil Aviation Policy (NCAP), 2016

- **Goal:** To expand and sustain growth in the aviation sector.
- **Key Components:**
 - Focuses on **regional connectivity** and making air travel affordable.
 - Allows **100% Foreign Direct Investment (FDI)** in aviation.
 - Sets targets for increasing both domestic and international air passenger numbers.
- **Impact:** Improved infrastructure, enhanced foreign investment, and increased connectivity for the common man.

Airport Infrastructure Development

- **Greenfield Airports:** New airports are being built in places like Navi Mumbai, Noida (Jewar), and Mopa (Goa) to handle growing air traffic.
- **Expansion of Existing Airports:** Airports in Delhi, Mumbai, and Bangalore have undergone expansion to manage increasing passenger numbers.
- **Public-Private Partnerships (PPP):** Private players are involved in developing and managing airports, enhancing efficiency and customer service.

Air Navigation and Safety Improvements

- **GAGAN (GPS-Aided Geo-Augmented Navigation):** Provides satellite-based navigation and improved safety for flights over India.
- **ASMGCS (Advanced Surface Movement Guidance and Control System):** Assists in managing aircraft movement on the ground at busy airports, reducing delays and enhancing safety.

Major Airports in India





- **Indira Gandhi International Airport (Delhi):** The busiest airport in India, handling significant domestic and international traffic.
- **Chhatrapati Shivaji Maharaj International Airport (Mumbai):** A major hub for domestic and international flights.
- **Kempegowda International Airport (Bangalore):** A rapidly growing airport with high passenger traffic and a hub for tech city Bangalore.

Pipelines in India

Pipelines are essential for transporting fuels like natural gas, crude oil, and petroleum products across the country. They are cost-effective, safe, and critical for meeting India's energy needs.

Importance of Pipelines

- **Efficient Transport:** Pipelines are economical for transporting bulk energy products over long distances.
- **Safety:** Reduces the risk of accidents and spills compared to road and rail transport.
- **Energy Access:** Ensures a steady supply of fuels to regions that may not have access to other transport networks.

Types of Pipelines in India

- **Crude Oil Pipelines:** Transport crude oil from ports or oil fields to refineries.
- **Petroleum Product Pipelines:** Carry refined products like petrol, diesel, and kerosene from refineries to storage depots and consumer points.
- **Natural Gas Pipelines:** Transport natural gas from production fields to consumers, including industries, households, and power plants.

Key Pipeline Networks

Hajira-Bijapur-Jagdishpur (HBJ) Pipeline

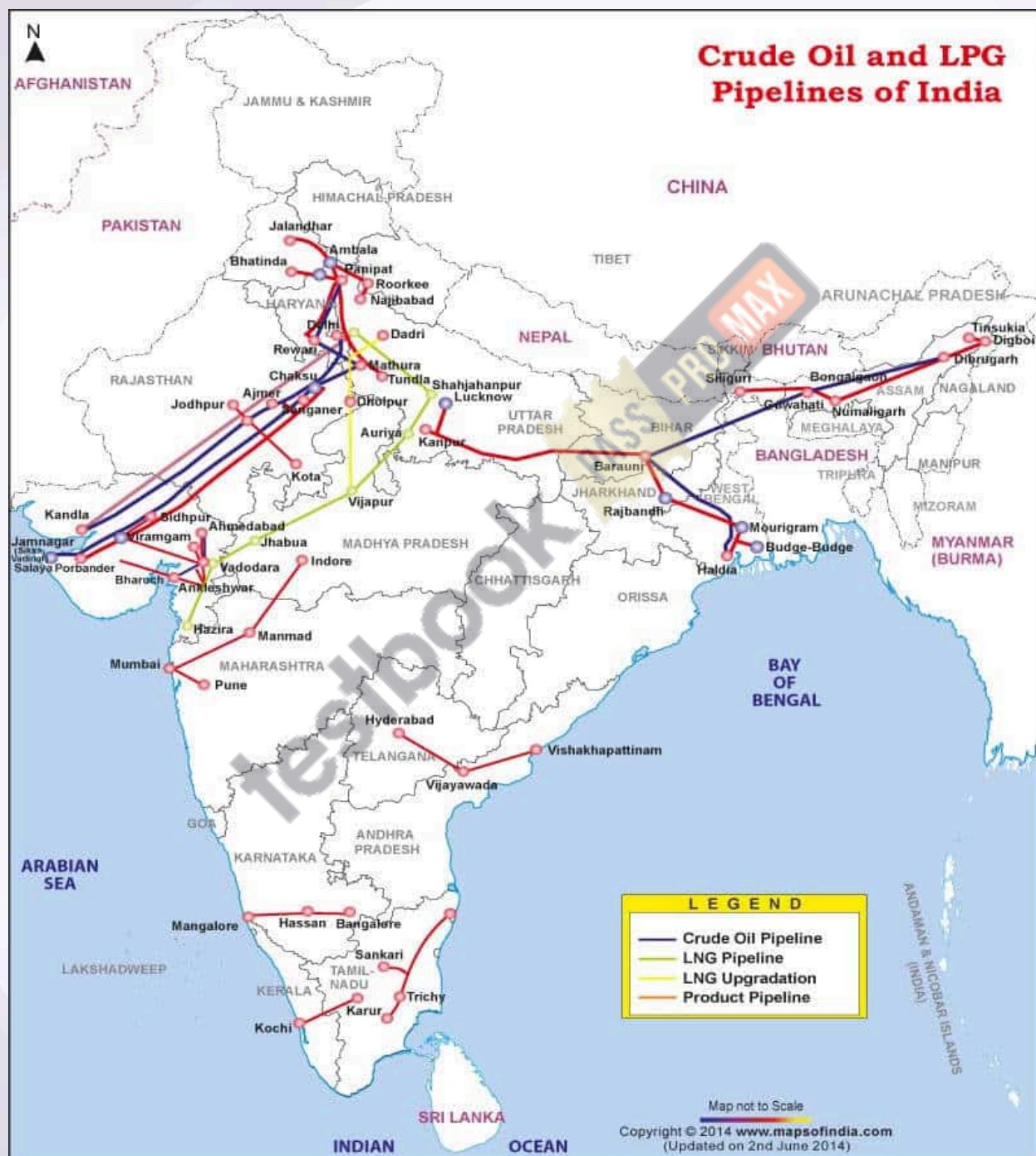
- **Description:** India's first and longest natural gas pipeline, spanning 1,750 km.
- **Route:** From western India (Hajira in Gujarat) to northern states like UP and Bihar.
- **Significance:** Supports industries and fertilizing plants with a steady gas supply, and powers major industrial zones.

Kandla-Bhatinda Pipeline

- **Purpose:** Transport petroleum products from Kandla (Gujarat) to Bhatinda (Punjab).
- **Impact:** Ensures reliable fuel supply in the northern regions and reduces transportation costs.

Jamnagar-Loni LPG Pipeline

- **Claim to Fame:** Longest LPG pipeline in the world.
- **Route:** Carries LPG from Jamnagar (Gujarat) to Loni (Uttar Pradesh).
- **Significance:** Supports domestic fuel needs across North India, providing a steady supply of LPG.



Eastern India Pipeline Network

- Developed by **GAIL** (Gas Authority of India Ltd.) to support gas distribution across Eastern India.
- Covers states like West Bengal, Bihar, Odisha, and Jharkhand, supplying gas for industrial and household use.

Government Initiatives in Pipeline Infrastructure

National Gas Grid

- **Objective:** To create a nationwide gas pipeline network that ensures reliable and widespread availability of natural gas.
- **Key Projects:**
 - Linking various pipeline networks across the country to form a unified grid.
 - Extending coverage to Northeast India, Jammu & Kashmir, and remote areas.
- **Impact:** Increases accessibility to natural gas, providing a cleaner energy option for industries and households.

Pradhan Mantri Urja Ganga Project

- **Objective:** To bring natural gas to eastern and northeastern states, including UP, Bihar, Jharkhand, Odisha, and West Bengal.
- **Scope:** Involves laying over 3,000 km of pipeline.
- **City Gas Distribution (CGD):** Establishes piped gas and CNG networks in cities under this route.
- **Benefits:** Encourages a shift to clean energy, boosts industrialization in eastern India, and reduces pollution.

City Gas Distribution (CGD) Networks

- **Goal:** To provide piped natural gas (PNG) for households and CNG for vehicles in urban and semi-urban areas.
- **Impact:** Helps in reducing pollution from LPG cylinders and petrol/diesel vehicles.
- **Coverage:** Major cities and smaller towns are being connected to piped natural gas, supporting environmental sustainability.

Petroleum and Natural Gas Regulatory Board (PNGRB)

- **Objective:** Regulates pipeline infrastructure development and ensures fair competition.
- **Role:**
 - Oversees licensing, tariff regulations, and pipeline expansions.
 - Aims to provide transparency in the gas sector and ensures consistent supply and quality.